TRYOLABS IT ADMIN PROVISIONING SPECIFICATION • RETAIL EDITION (RETAIL PULSE)
Delivered by Tryolabs • Powered by Google Antigravity & Gemini Enterprise

IT Administrator Workstation Requirements — Retail Edition (Retail Pulse)

Complete OS, CLI Binary, Python Library (`requirements.txt`), Port, EDR & File Permission Specification for Non-Admin Computers (`antigravity-workshop-retail`)

CRITICAL REQUIREMENT: ATTENDEES DO NOT HAVE INSTALL PERMISSIONS
Workshop attendees will use locked-down corporate computers without local administrator (**`sudo`** / Administrator) privileges. IT Administrators must pre-install the system binaries below and pre-build the **`antigravity-workshop-retail/.venv`** virtual environment using **`admin_provision_workstation.sh`** (macOS/Linux) or **`admin_provision_workstation.ps1`** (Windows), then grant recursive read/write ownership of the **`antigravity-workshop-retail`** directory to the standard attendee user account.

1. Hardware, OS, Local Loopback Ports & EDR/Firewall Rules

Category	Requirement Specification
Operating System	macOS 13+ (Ventura/Sonoma/Sequoia), Windows 10/11 (64-bit) with Git Bash or WSL2, or Linux (Ubuntu 22.04+, Debian 12+, RHEL 9+)
CPU & Memory (RAM)	Minimum 4 vCPU / 8 GB RAM (8+ vCPU / 16 GB RAM recommended); 10 GB free disk space
Local TCP Ports (Loopback Only)	Retail Pulse Command Center UI (**`127.0.0.1:8002`**) and **`127.0.0.1:8080`** (Google ADK Dev UI **`adk web adk_agents/retail_pulse`**) — Unprivileged local binding only
EDR / Antivirus Subprocess Rule	Allow **`.venv/bin/python`** to run as a child process over **`stdio`** inside **`antigravity-workshop-retail/`** for FastMCP (**`mcp_server/retail_mcp.py`**) and **`PreToolUse`** hooks (**`scripts/audit_hook.py`**)
Outbound HTTPS Allowlist (443)	**`*.googleapis.com`** (**`cloudaicompanion`**, **`aiplatform`**, **`discoveryengine`**, **`run`**), **`accounts.google.com`**, **`oauth2.googleapis.com`**

2. Complete System Software & CLI Binary Matrix (Must Be in System PATH)

Software / Binary	Version	Verify Command	Role in Retail Edition (Retail Pulse)	Status
1. Google Antigravity	Latest Enterprise	**`agy --version`** / App Launch	Antigravity 2.0 Desktop App, Antigravity IDE, and **`agy`** CLI in PATH	REQUIRED
2. Python 3 Runtime	>= 3.11 (>= 3.10 ok)	**`python3 --version`**	Must include standard modules **`venv`**, **`pip`**, **`sqlite3`**, **`decimal`**	REQUIRED
3. Astral **`uv`**	>= 0.5.0	**`uv --version`**	Fast Python environment & package manager in system PATH	REQUIRED
4. Git CLI	>= 2.39	**`git --version`**	Cloning **`antigravity-workshop-retail`** and inspecting diffs	REQUIRED
5. SQLite 3 CLI	>= 3.37	**`sqlite3 --version`**	Lab 1 & Lab 3 database inspection (**`data/retail_pulse.db`**) and hook tests	REQUIRED
6. Bash & cURL	bash >= 3.2, curl	**`bash --version`**	Executes **`./install.sh`**, **`./verify.sh`**, and local A2A checks	REQUIRED
7. Terraform CLI	>= 1.5.0	**`terraform -version`**	Lab 4 Cloud Run & IAM validation (**`infra/terraform/`**)	REQUIRED (Lab 4)
8. Google Cloud SDK	>= 460.0.0	**`gcloud --version`**	Gemini Enterprise auth & Lab 4 deployment (**`gcloud auth`**)	REQUIRED (Lab 4)
9. Chromium Browser	Chrome / Edge	Open browser	Viewing **`:8002`** dashboard, **`:8080 adk web`**, & HTML slides	REQUIRED

3. Complete Python Library Specification (`antigravity-workshop-retail/requirements.txt`)

All Python libraries below must be pre-installed by the IT Administrator inside **`antigravity-workshop-retail/.venv`** so that non-admin attendees never need to run **`pip install`** during the workshop:

Python Package	Min Version	Locked Version	Usage Across Retail Edition (Retail Pulse) Labs 1–4
`fastapi`	>=0.110.0	0.141.1	Core REST API & A2A JSON-RPC 2.0 endpoint (**`/api/v1/a2a`**) across Labs 1–4
`uvicorn`	>=0.27.0	0.54.0	Local ASGI server for the live dashboard on **`127.0.0.1:8002`**
`pydantic`	>=2.6.0	2.14.1	Strict schema validation for GS1 GTIN-14 SKUs, WMS inventory nodes, promo rules, OTIF contracts, and A2A payloads

Python Package	Min Version	Locked Version	Usage Across Retail Edition (Retail Pulse) Labs 1–4
`pytest`	>=8.0.0	9.1.1	Automated verification suite executed by **`./verify.sh`** across Labs 1, 2, 3, and 4
`httpx`	>=0.27.0	0.28.1	FastAPI **`TestClient`** transport and async A2A protocol requests
`mcp`	>=1.2.0	2.3.0	Official Model Context Protocol SDK (**`FastMCP`**) for Lab 3 (**`mcp_server/retail_mcp.py`**)
`google-adk`	>=1.0.0	2.12.0	Google Agent Development Kit (**`LlmAgent`**, **`Runner`**, and **`adk web`** for **`adk_agents/retail_pulse`** on **`:8080`**)
`google-genai`	>=1.0.0	2.33.0	Google GenAI / Vertex AI client SDK used by ADK multi-agent hierarchies
`google-auth`	>=2.28.0	2.62.0	Google Cloud IAM & Application Default Credentials authentication
`aiosqlite`	>=0.20.0	0.22.1	Async SQLite session storage backend required by **`adk web`**
`sse-starlette` & **`starlette`**	>=2.0.0 / >=0.36.0	3.5.0 / 1.9.0	ASGI routing and Server-Sent Events streaming for FastMCP and ADK
`websockets`	>=12.0	15.0.1	Real-time bidirectional event streaming in **`adk web`**
`python-dotenv` & **`PyYAML`**	>=1.0.0 / >=6.0.1	1.2.4 / 6.0.3	Environment configuration and YAML frontmatter parsing for Lab 2 **`SKILL.md`**
`python-multipart` & **`requests`**	>=0.0.9 / >=2.31.0	0.0.32 / 2.34.2	FastAPI form handling and HTTP utilities
`tenacity`, **`watchdog`**, **`graphviz`**	>=8.2 / >=4.0 / >=0.20	9.1.4 / 6.0.0 / 0.21	Agent retry resilience, file watching, and ADK agent graph rendering
`opentelemetry-api` & **`sdk`**	>=1.24.0	1.42.1	Multi-agent execution trace telemetry inspected in **`adk web`**

4. Exact `requirements.txt` for Retail Edition (Retail Pulse)

Use the included **`requirements.txt`** (and **`requirements-lock.txt`** for air-gapped wheelhouses) located in **`antigravity-workshop-retail/requirements.txt`**:

```
# requirements.txt — Tryolabs Google Antigravity Workshop: Retail Edition (Retail Pulse)
fastapi>=0.110.0
uvicorn>=0.27.0
starlette>=0.36.0
pydantic>=2.6.0
python-multipart>=0.0.9
python-dotenv>=1.0.0
pytest>=8.0.0
httpx>=0.27.0
anyio>=4.0.0
requests>=2.31.0
mcp>=1.2.0
sse-starlette>=2.0.0
google-adk>=1.0.0
google-genai>=1.0.0
google-auth>=2.28.0
aiosqlite>=0.20.0
websockets>=12.0
PyYAML>=6.0.1
tenacity>=8.2.0
watchdog>=4.0.0
graphviz>=0.20
opentelemetry-api>=1.24.0
opentelemetry-sdk>=1.24.0
```

5. Critical File Permissions for Non-Admin Attendees & One-Step Provisioning

Because attendees run under unprivileged OS user accounts, **the attendee's user account MUST have full read/write/modify permissions** on the entire **`antigravity-workshop-retail`** folder (**`.venv/`**, **`data/retail_pulse.db`**, **`.agents/`**, **`src/`**, **`mcp_server/retail_mcp.py`**, **`adk_agents/retail_pulse/`**, **`labs/`**). Run our automated provisioning script as Admin to build **`.venv`**, seed **`data/retail_pulse.db`**, and set user ownership automatically:

```
# Option A: Automated IT Admin Provisioning & Ownership Assignment (macOS / Linux)
cd antigravity-workshop-retail
sudo ./admin_requirements/admin_provision_workstation.sh --user <attendee_username>

# Option B: Automated IT Admin Provisioning & ACL Assignment (Windows PowerShell as Admin)
cd antigravity-workshop-retail
.\admin_requirements\admin_provision_workstation.ps1 -AttendeeUser "CORP\<attendee_username>"

# Option C: Verify Readiness as the Unprivileged Attendee User (Zero-Install Mode)
./admin_requirements/admin_provision_workstation.sh --verify-only
```

6. Google Cloud APIs, IAM Roles & Permissions (Antigravity + Gemini Enterprise)

To use **Google Antigravity** (**`agy`**, Antigravity IDE, and Antigravity 2.0) with **Gemini Enterprise**, execute **Google ADK (`adk web`)** against **Vertex AI**, and deploy/register the **`retail-pulse`** A2A agent in Lab 4, the Cloud Administrator must enable the APIs and assign the IAM roles below on the workshop Google Cloud project (**`<WORKSHOP_GCP_PROJECT_ID>`**).

6.1 Required Google Cloud APIs to Enable (`gcloud services enable`)

Google Cloud API Service	API Title	Purpose in Retail Edition (Retail Pulse)	Status
`cloudaicompanion.googleapis.com`	Gemini for Google Cloud API	Core backend API for Google Antigravity (**`agy`**, IDE, 2.0) under Gemini Enterprise subscriptions, code generation, agentic tool execution, and seat entitlement verification	REQUIRED (All Labs)
`aiplatform.googleapis.com`	Vertex AI API	Powers Google ADK (**`adk web adk_agents/retail_pulse`**) and **`google-genai`** SDK (**`GOOGLE_GENAI_USE_VERTEXAI=TRUE`**) for Lab 4 multi-agent execution	REQUIRED (Lab 4)
`discoveryengine.googleapis.com`	Gemini Enterprise / Discovery Engine API	Registers and invokes the **`retail-pulse`** A2A (**`AgentCard`**) endpoint inside Gemini Enterprise (**`register_gemini_enterprise_agent.sh`**)	REQUIRED (Lab 4)
`serviceusage.googleapis.com` & **`cloudresourcemanager.googleapis.com`**	Service Usage & Resource Manager APIs	Quota project validation (**`x-goog-user-project`**), API state verification, and Terraform provider initialization	REQUIRED (All Labs)
`run.googleapis.com`, **`artifactregistry.googleapis.com`**, **`cloudbuild.googleapis.com`**, **`iam.googleapis.com`**, **`logging.googleapis.com`**	Cloud Run, Artifact Registry, Cloud Build, IAM & Logging APIs	Deploys the containerized **`retail-pulse`** FastAPI + ADK + A2A service (**`infra/terraform/`**) and records **`cloudaicompanion.googleapis.com`** Cloud Audit Logs	REQUIRED (Lab 4 Deploy)

6.2 Required IAM Roles & Permissions (Attendee Accounts & Cloud Run Runtime)

In addition to assigning an active **Gemini Enterprise / Gemini Code Assist Enterprise license seat** to each attendee in **Google Cloud Console -> Admin for Gemini -> License Management**, grant the following IAM roles:

IAM Role ID	Assigned Principal	Key Included IAM Permissions	Workshop Requirement
`roles/cloudaicompanion.user`	Every Attendee (`user:<attendee>@company.com`)	`cloudaicompanion.companions.generateChat`, `cloudaicompanion.companions.generateCode`, `cloudaicompanion.instances.completeCode`, `cloudaicompanion.instances.generateChat`, `cloudaicompanion.instances.generateCode`, `cloudaicompanion.instances.queryEffectiveSetting`, `cloudaicompanion.entitlements.get`	Mandatory to sign in to Google Antigravity (`agy`, IDE, 2.0) and consume a Gemini Enterprise seat
`roles/serviceusage.serviceUsageConsumer`	Every Attendee (`user:<attendee>@company.com`)	`serviceusage.services.use`, `serviceusage.services.get`, `serviceusage.services.list`	Mandatory to bill API quota calls against the workshop GCP project (`gcloud auth application-default set-quota-project`)
`roles/aiplatform.user`	Every Attendee & `retail-pulse-sa` Service Account	`aiplatform.endpoints.predict`, `aiplatform.endpoints.generateContent`, `aiplatform.endpoints.streamGenerateContent`, `aiplatform.models.get`	Mandatory for Lab 4 ADK (`adk web adk_agents/retail_pulse`) and multi-agent Vertex AI execution
`roles/discoveryengine.editor` (or `roles/discoveryengine.viewer`)	Attendee / Instructor (Lab 4)	`discoveryengine.engines.get`, `discoveryengine.engines.list`, `discoveryengine.assistants.get`, `discoveryengine.agents.create`, `discoveryengine.agents.update`	Required in Lab 4 to register `retail-pulse` (`/.well-known/agent.json`) with Gemini Enterprise
`roles/cloudaicompanion.admin` & `roles/logging.privateLogViewer`	Cloud Admin / Workshop Instructor	`cloudaicompanion.entitlements.*`, `cloudaicompanion.instances.*`, `logging.logEntries.list`	Used by Admin to assign Gemini Enterprise license seats and by Instructor to demo Cloud Audit Logs
`roles/run.admin`, `roles/artifactregistry.admin`, `roles/cloudbuild.builds.editor`, `roles/iam.serviceAccountAdmin`, `roles/iam.serviceAccountUser`	Instructor / Terraform Deployer (Lab 4)	`run.services.create`, `artifactregistry.repositories.create`, `cloudbuild.builds.create`, `iam.serviceAccounts.actAs`	Required to run `terraform apply` (`infra/terraform/`) and deploy `retail-pulse` to Cloud Run v2

6.3 One-Step `gcloud` Commands to Enable APIs & Grant Attendee Roles

```
export PROJECT_ID="<WORKSHOP_GCP_PROJECT_ID>"
gcloud services enable cloudaicompanion.googleapis.com aiplatform.googleapis.com \
  discoveryengine.googleapis.com serviceusage.googleapis.com \
  cloudresourcemanager.googleapis.com run.googleapis.com \
  artifactregistry.googleapis.com cloudbuild.googleapis.com \
  iam.googleapis.com logging.googleapis.com --project="${PROJECT_ID}"

for ROLE in roles/cloudaicompanion.user roles/serviceusage.serviceUsageConsumer \
  roles/aiplatform.user roles/discoveryengine.editor roles/viewer; do
  gcloud projects add-iam-policy-binding "${PROJECT_ID}" \
    --member="user:<ATTENDEE_EMAIL>" --role="${ROLE}"
done
```

7. Pre-Delivery IT Admin Verification Checklist

#	Check	Pass Criteria (Run as Standard Non-Admin Attendee User)
1	System Binaries in PATH	**`python3`**, **`uv`**, **`git`**, **`sqlite3`**, **`bash`**, **`curl`**, **`terraform`**, **`gcloud`**, and **`agy`** resolve in standard user PATH
2	Pre-Built **`.venv`** Ready	**`./.venv/bin/python -c 'import fastapi, uvicorn, pydantic, pytest, httpx, mcp, google.adk, google.genai'`** exits 0
3	ADK Dev UI Binary Ready	**`./.venv/bin/adk --version`** and **`./scripts/run_adk_web.sh --check`** (verifying **`adk_agents/retail_pulse`**) exit 0
4	Zero-Install Mode Active	Running **`./install.sh`** as standard user skips network downloads (**`[OK] Pre-provisioned .venv detected`**) and seeds **`data/retail_pulse.db`**
5	Non-Admin Write Access	Standard attendee user can write to **`data/`**, **`.agents/`**, **`src/`**, and **`mcp_server/retail_mcp.py`** without **`Permission denied`** errors
6	GCP IAM & APIs Ready	**`cloudaicompanion.googleapis.com`** & **`aiplatform.googleapis.com`** enabled; attendee has **`roles/cloudaicompanion.user`** & **`roles/serviceusage.serviceUsageConsumer`**